Curriculum Vitae

Dr. Julien P. Irmer

Postdoctoral Researcher

University of Freiburg

(One Child, ⋆ 2023)

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Research Profile

I develop statistical methods for complex psychological data and study their performance from theoretical derivation to practical application. My research spans mathematical methodology, simulation-based evaluation, open-source software development, and methodological guidance for empirical research, aiming to improve reproducibility, causal inference, and the quality of scientific evidence.

Areas of Expertise: Statistical Methodology · Longitudinal Data Analysis · Causal Inference · Structural Equation Modeling · Simulation-Based Evaluation · Machine Learning · Open Science · AI for Scientific Research

Academic Positions

Albert-Ludwigs-Universität Freiburg

Since 10.2025 **Research Associate**, **Postdoctoral Researcher**, Evaluation Lab. Research focus: Statistical methodology, causal inference, longitudinal modeling, continuous-time modeling, structural equation modeling, simulation studies, machine learning, meta-analysis, nonparametric methods, power analysis. **Head:** Prof. Dr. Manuel C. Voelkle.

Humboldt-Universität zu Berlin

04.2024–09.2025 **Research Associate**, **Postdoctoral Researcher**, Psychological Methods Lab. **Head:** Prof. Dr. Manuel C. Voelkle.

Goethe University Frankfurt

04.2019–04.2024 **Research Associate, Doctoral Researcher**, Methodology Division. Research focus: Nonlinear latent variable modeling, non-parametric estimation, model selection, simulation-based power estimation, and adaptive algorithms. **Head:** Prof. Dr. Andreas G. Klein.

2017–2019 **Student Research Assistant**, Methodology Division. Research focus: Quasi-maximum likelihood methods and model selection. **Supervisor:** Dr. Rebecca D. Büchner.

2017–2018 **Research Intern**, Neuro Lab. Research focus: EEG data analysis and the influence of filter techniques on EEG modeling.

2015–2019 **Student Teaching Assistant**, Statistics in Mathematics Lab and Methodology Division in Psychology. Teaching focus: Statistics in mathematics, psychology, environmental sciences, and biology, as well as programming with R and Python.

2012–2016 **Student Assistant**, Work and Organizational Psychology Lab. Focus: Data collection for meta-analyses.

Education

Goethe University Frankfurt

04.2019–09.2024 **Ph.D. in Psychological Methods** — Dr. rer. nat. *(summa cum laude,* excellent). Dissertation: *Model Selection and the Estimation of Statistical Power in Nonlinear Structural Equation Modeling.* https://doi.org/10.21248/gups.87712

2017–04.2019 **Master of Science: Mathematics** (Grade: 1.0). Thesis: *Linear and Quadratic Structural Equation Modeling – Studying the Unobservable.*

2016–09.2021 **Master of Science: Psychology** (Grade: 1.0). Thesis: *When Can Ordered Categorical Variables Be Treated as Continuous in Quadratic and Interaction Structural Equation Modeling?*

2011–2017 **Bachelor of Science: Mathematics.** Thesis: *Do Things Change? - An Analysis of the FDpV (Filtered Derivative with p-Value Method).*

2011–2016 **Bachelor of Science: Psychology.** Thesis: *The Instrument for Stress Oriented Job Analysis (ISTA) – A Meta-Analytical Validation Using the Challenge Stressor-Hindrance Stressor Concept and the Job Demands-Resources Model.*

Adelaide High School, South Australia

2009–2011 **South Australian Certificate of Education.** (Translated grade: 1.0).

Research Output

* Shared first authorship.

Peer-Reviewed Articles

In press

11) Sperl, M. F. J., Baumgärtner, L., Bach, K. M., Bamberg, C., Behlau, C., Bergmann, B., Bienefeld, M., Bleckmann, E., Danböck, S. K., Dreston, J. H., Eckardt, V. C., Frick, S., Friehs, M.-T., Handke, L., Hein, I., Hutmacher, F., **Irmer, J. P.,** Kause, A., Kern, M., … & Neef, N. E. (in press). Künstliche Intelligenz bei Abschlussarbeiten, Dissertationen und Habilitationsschriften in der Psychologie – Eine Perspektive der DGPs-Jungmitglieder und Psychologie-Studierenden [Artificial intelligence in bachelor's, master's, doctoral, and habilitation theses in psychology: Perspectives of early career members of the German Psychological Society (DGPs) and Psychology students]. *Psychologische Rundschau.*

Published

10) Nestler, S., Humberg, S., Debelak, R., Heck, D. W., Henninger, M., Voelkle, M., Frick, S., **Irmer, J. P.,** Scharf, F., & Frey, A. (2026). Automating the Scientist? Perspektiven der Fachgruppe Methoden und Evaluation auf die Nutzung von KI in der psychologischen Forschung. [Perspectives of the methods and evaluation section on the use of AI in psychological research] *Psychologische Rundschau. 77* (3), 129–135. https://doi.org/10.1026/0033-3042/a000763

9) **Irmer, J. P.**, Schermelleh-Engel, K., & Schmidt, P. (2026). Editorial: Introduction to the special issue "Methodological challenges of complex latent mediator and moderator models". *Behavior Research Methods, 58* (127), 1–9. 10.3758/s13428-026-02985-3

8) Wallot, S.*, **Irmer, J. P.***, Tschense, M., Kuznetsov, N., Højlund, A., & Dietz, M. (2026). A multivariate method for dynamic system analysis: Multivariate detrended fluctuation analysis using generalized variance. *Topics in Cognitive Science, 18* (3), e12688. 10.1111/tops.12688

7) Escaffi-Schwarz, M., Gempp, R., & **Irmer, J. P.** (2025). Meta-analyzing the factor structure and reliability of measurement instruments: An R-based tutorial. *International Journal of Psychology, 60* (2), 1–9. 10.1002/ijop.70003

6) **Irmer, J. P.**, Klein, A. G., & Schermelleh-Engel, K. (2024). Model-implied simulation-based power estimation for correctly specified and distributionally misspecified models: Applications to nonlinear and linear structural equation models. *Behavior Research Methods, 56,* 8897–8931. 10.3758/s13428-024-02507-z

5) **Irmer, J. P.**, Klein, A. G., & Schermelleh-Engel, K. (2024). Estimating power in complex nonlinear structural equation modeling including moderation effects: The powerNLSEM R-package. *Behavior Research Methods, 56,* 8897–8931. 10.3758/s13428-024-02476-3

3) Grønneberg, S.*, & **Irmer, J. P.*** (2024). Non-parametric regression among factor scores: Motivation and diagnostics for nonlinear structural equation models. *Psychometrika, 89* (3), 822–850. 10.1007/s11336-024-09959-4

3) Holman, D., Escaffi-Schwarz, M., Vasquez, C. A., **Irmer, J. P.**, & Zapf, D. (2024). Does job crafting affect employee outcomes via job characteristics? A meta-analytic test of a key job crafting mechanism. *Journal of Occupational and Organizational Psychology, 97* (1), 47–73. 10.1111/joop.12450

2) Ehmann, P., Beavan, A., Spielmann, J., Mayer, J., Altmann, S., Ruf, L., Rohrmann, S., **Irmer, J. P.**, & Englert, E. (2022). Perceptual-cognitive performance of youth soccer players in a 360°-environment: Differences between age groups and performance levels. *Psychology of Sport & Exercise, 59,* 102120. 10.1016/j.psychsport.2021.102120

1) **Irmer, J. P.**, Kern, M., Schermelleh-Engel, K., Semmer, N. K., & Zapf, D. (2019). ISTA – The instrument for stress oriented job analysis: A meta-analysis. *Zeitschrift für Arbeits- & Organisationspsychologie [German Journal of Work and Organizational Psychology], 63* (4), 217–237. 10.1026/0932-4089/a000312

Preprints

3) Eisenberg, M. J., Schultze, M., **Irmer, J. P.**, Draschkow, D., & Kumle, L. (2026). *Power analysis for mixed-models: A model-implied simulation-based power estimation approach* [Preprint]. OSF. https://osf.io/b5x9e
2) Vink, P. A., Mulder, J. D., **Irmer, J. P.**, & Hamaker, E. L. (2026). *How to account for unobserved baseline confounders with time-invariant and time-varying effects in cross-lagged panel research* [Preprint; manuscript submitted for publication]. PsyArXiv. https://osf.io/preprints/psyarxiv/sdjry_v2
1) Uyar, A., **Irmer, J. P.**, Repantis, D., Wolff, M., Lueken, U., & Evens, R. (2026). *Correlates of post-traumatic and perceptual symptoms following challenging psychedelic experiences: A cross-sectional study* [Preprint; manuscript submitted for publication]. ResearchSquare.

Books and Book Chapters

2) Schermelleh-Engel, K., Gäde, J. C., & **Irmer, J. P.** (2021). R-Syntax – direkte Programmierung zu Kapitel 14: Klassische Methoden der Reliabilitätsschätzung. Zusatzmaterialien zu H. Moosbrugger & A. Kelava (Hrsg.), *Testtheorie und Fragebogenkonstruktion* (3., vollständig neu bearbeitete, erweiterte und aktualisierte Auflage) [R syntax – programming implementation of chapter 14: Classical methods of reliability estimation. Supplementary materials to H. Moosbrugger & A. Kelava (eds.), *Test Theory and Questionnaire Construction* (3rd, revised and updated)]. Heidelberg: Springer. http://www.lehrbuch-psychologie.springer.com
1) Schermelleh-Engel, K., Gäde, J. C., & **Irmer, J. P.** (2021). Mplus-Syntax – direkte Programmierung zu Kapitel 14: Klassische Methoden der Reliabilitätsschätzung. Zusatzmaterialien zu H. Moosbrugger & A. Kelava (Hrsg.), *Testtheorie und Fragebogenkonstruktion* (3., vollständig neu bearbeitete, erweiterte und aktualisierte Auflage) [Mplus syntax – programming implementation of chapter 14: Classical methods of reliability estimation. Supplementary materials to H. Moosbrugger & A. Kelava (eds.), *Test Theory and Questionnaire Construction* (3rd, revised and updated)]. Heidelberg: Springer. http://www.lehrbuch-psychologie.springer.com

Commentaries and Other Publications

1) Sperl, M. F. J., Attig, C., Baumgärtner, L., Behlau, C., Bergmann, B., Bienefeld, M., Biller, A. M., Brandt, N. D., Bühler, J. L., Frauhammer, L. T., Frick, S., Friehs, M.-T., Frischen, U., Handke, L., Hartmann, H., Hösterey, S., Hutmacher, F., Iffland, J. A., **Irmer, J. P.**, … Wendt, A. N. (2025). Gute Betreuung, faire Bewertungen und angemessene Arbeitsbedingungen ermöglichen exzellente Promotionen in der Psychologie: Kommentar der Jungmitgliedervertretung der DGPs zum Positionspapier. *Psychologische Rundschau, 76* (3), 196–199. https://doi.org/10.1026/0033-3042/a000729

Research Software

6) **Irmer, J. P.** (2026). *glmLink: Models for Power* (Version 0.5.0, development branch) [R package]. GitHub repository. https://github.com/jpirmer/glmLink/tree/dev
5) Arnold, M., & **Irmer, J. P.** (2026). *ipcr: Individual Parameter Contribution Regression* (Version 0.3.3, development branch) [R package]. GitHub repository. https://github.com/manuelarnold/ipcr/tree/dev2
4) **Irmer, J. P.** (2024). *powerNLSEM: Model-implied simulation-based power estimation (MSPE) for nonlinear structural equation models* (Version 0.1.2) [R package]. https://doi.org/10.32614/CRAN.package.powerNLSEM
3) **Irmer, J. P.**, & Wallot, S. (2023). *mvDFA: Multivariate Detrended Fluctuation Analysis* (Version 0.0.4) [R package]. https://doi.org/10.32614/CRAN.package.mvDFA

2) Schultze, M., **Irmer, J. P.**, & Nehler, K. J. (Eds.). (2022). *PandaR: Praktische Anwendungen der Datenanalyse in R* [Practical ApplicatioNs of Data Analysis in R]. https://pandar.netlify.app/

1) Schmalbach, B., **Irmer, J. P.**, & Schultze, M. (2019). *ezCutoffs: Fit measure cutoffs in SEM* (Version 1.0.1) [R package]. https://doi.org/10.13140/RG.2.2.16315.77600

Hackathon

2026 — **Multi-Analyst Study and Hackathon on Interpretable Machine Learning for Longitudinal and Clustered Data**, TU Dortmund University, Dortmund, Germany. Accepted contribution: *An Interpretable Machine Learning Approach to Density-Ratio Weighting Under Clustered Data: Application to PIRLS 2016–2021.* Presented the methodological approach and preliminary results. Manuscript in preparation.

Funding and Awards

Awards and Honors

2025 — **Gustav A. Lienert Award** — Award for the best dissertation submitted between 2023 and 2025, presented by the Methods and Evaluation Section of the German Psychological Society (DGPs). **Award amount:** €750.

2025 — **Award for the Best Dissertation in the Natural Sciences** — Awarded by Goethe University Frankfurt and sponsored by the Verein der Freunde und Förderer der Goethe-Universität Frankfurt e.V. **Award amount:** €10,000.

2025 — **Top Cited Article Recognition** — Co-author of *Does Job Crafting Affect Employee Outcomes via Job Characteristics? A Meta-Analytic Test of a Key Job Crafting Mechanism,* recognized as one of the most-cited papers published in the *Journal of Occupational and Organizational Psychology* in 2023.

2023 — **OER Prize** — Awarded by HessenHub for the Statistics I course developed with the PandaR team, including Martin Schultze, Kai Nehler, and colleagues. **Award amount:** €500.

2021 — **Young YAVIS Award** — Awarded by the Institute of Psychology at Goethe University Frankfurt for outstanding teaching. **Award amount:** €100.

2019 — **Award for the Best Master's Degree** — Awarded for the best master's degree of the 2018/2019 academic year at the Institute of Mathematics, Goethe University Frankfurt. Sponsored by Alte Leipziger Lebensversicherung a. G. and the Association for the Promotion of Mathematics at Goethe University Frankfurt e.V. **Award amount:** €400.

2017–2019 — **Deutschlandstipendium** — Scholarship for talented and high-achieving students at Goethe University Frankfurt, Mathematics in two consecutive years. **Award amount:** €5,400.

2010 — **Certificate of Outstanding Academic Achievement** — Awarded for outstanding achievement in mathematical studies at Adelaide High School.

Conference Funding

2025 — **Conference Travel Grant**, Methods and Evaluation Section of the German Psychological Society (DGPs), for participation in the section conference. **Award amount:** €250.

2023 **Conference Travel Grant**, Dean's Office, Institute of Psychology, Goethe University Frankfurt, for participation in the conference of the Methods and Evaluation Section of the German Psychological Society. **Award amount:** €110.

2023 **Conference Travel Grant**, Dean's Office, Institute of Psychology, Goethe University Frankfurt, for participation in the *Advanced Techniques for Longitudinal Data Analysis in Social Science* conference. **Award amount:** €200.

2020 **Conference Travel Grant**, Dean's Office, Institute of Psychology, Goethe University Frankfurt, for participation in the Structural Equation Modeling Working Group Meeting in Vienna. The conference was subsequently canceled because of the COVID-19 pandemic. **Award amount:** €128.

Talks and Conference Contributions

Invited Talks

10) **Irmer, J. P.** (2026, March 26). From nonparametric model selection to model-implied simulation-based power estimation: Towards an adaptive, machine-learning inspired framework for complex structural equation models and beyond [Invited talk]. Society & Choice Research Seminars, University of Basel, Switzerland.
9) **Irmer, J. P.** (2025, September). Model selection and the estimation of statistical power for nonlinear structural equation modeling [Award keynote]. 17th Conference of the Methods & Evaluation Section of the German Psychological Society (DGPs), Berlin, Germany.
8) **Irmer, J. P.** (2025, June 25). From nonparametric model selection to model-implied simulation-based power estimation: A machine-learning-inspired, adaptive simulation framework for complex structural equation models and beyond [Invited talk]. Dynamic Modeling Lab Meeting, Utrecht University, Netherlands.
7) **Irmer, J. P.** (2025, June 10). From model selection to a novel simulation-based power estimation method: Applications to nonlinear SEM [Invited talk]. Kolloquium Psychologische Methodenlehre, Philipps University Marburg, Germany.
6) **Irmer, J. P.** (2025, May 14). Model selection and the estimation of statistical power in nonlinear structural equation modeling [Invited presentation]. Presentation before the committee of the *Preis der Vereinigung der Freunde und Förderer der Johann Wolfgang Goethe-Universität e.V.,* Frankfurt, Germany.
5) **Irmer, J. P.** (2025, January). Overcoming the challenges of nonlinear SEM: A novel simulation-based power estimation method [Invited talk]. Department Colloquium, University of Kassel, Germany.
4) **Irmer, J. P.**, & Grønneberg, S. (2023, September). Motivation and diagnostics for nonlinear structural equation models using non-parametric regression among factor scores [Invited talk]. Research Seminar in Data Science, Norwegian Business School, Oslo, Norway.
3) **Irmer, J. P.** (2023, April). On problems and their solutions in nonlinear structural equation modeling [Invited talk]. Research Colloquium, Department of Psychological Methods, Philipps University Marburg, Germany.
2) **Irmer, J. P.** (2022, October). PandaR as an example for open teaching practices [Invited talk]. ReproducibiliTea Frankfurt, Goethe-University Frankfurt, Germany. Slides/OSF
1) **Irmer, J. P.** (2021, June). Nichtlineare Strukturgleichungsmodelle [Nonlinear structural equation modeling] [Invited talk]. Research Colloquium, Department of Psychological Methods, Humboldt-Universität zu Berlin, Germany.

Conference Presentations

22) Nehler, K. J., **Irmer, J. P.**, & Schultze, M. (2026, July). Why ordinal data can create spurious edges in psychological networks [Conference presentation]. *International Meeting of the Psychometric Society 2026 (IMPS 2026),* Seoul, Republic of Korea.

21) **Irmer, J. P.**, & Voelkle, M. C. (2025, September). No analytical power formula? No problem. A general model-implied simulation-based approach to power estimation for likelihood ratio tests with applications in complex SEM [Conference presentation]. 17th Conference of the Methods & Evaluation Section of the German Psychological Society (DGPs), Berlin, Germany.
20) **Irmer, J. P.** (2025, March). Model selection, diagnostics, and motivation for nonlinear structural equation models via non-parametric regression among factor scores [Conference presentation]. *DagStat 2025 – Joint Statistical Meeting of the DAGStat,* Berlin, Germany.
19) **Irmer, J. P.** (2025, March). Power up your nonlinear SEM analysis: Model-implied simulation-based power estimation with powerNLSEM using asymptotic normality [Conference presentation]. *Meeting of the Working Group Structural Equation Modeling,* Chemnitz, Germany.
18) **Irmer, J. P.** (2024, September). Estimating power in moderated mediation models: A model-implied simulation-based power estimation approach introducing the powerNLSEM R-package [Conference presentation]. *53rd Congress of the German Psychological Society (DGPS),* Vienna, Austria.
17) Hartig, J., Köhler, C., & **Irmer, J. P.** (2024, September). Sollte bei der Beurteilung des Itemfit anhand von Infit und Outfit auf die Verwendung pauschaler Cut-Off-Werte verzichtet werden? [Conference presentation]. *53rd Congress of the German Psychological Society (DGPS),* Vienna, Austria.
16) Schultze, M., **Irmer, J. P.**, & Nehler, K. J. (2023, September). pandaR [Conference presentation]. *CIHH23 Campus Innovation,* Hamburg, Germany.
15) **Irmer, J. P.**, Klein, A. G., & Schermelleh-Engel, K. (2023, September). Simulation-based semi-parametric estimation of power in nonlinear SEM [Conference presentation]. *15th Meeting of the DGPS Methods & Evaluation Division (FGME),* Konstanz, Germany.
14) Grønneberg, S., & **Irmer, J. P.** (2023, September). Non-parametric regression among factor scores [Conference presentation]. *Conference for Frontier Research in Educational Measurement (FREMO),* Oslo, Norway.
13) Kern, M., **Irmer, J. P.**, Dormann, C., & Zapf, D. (2023, September). Analyse komplexer Längsschnittdaten mit zeitkontinuierlichen Strukturgleichungsmodellen am Beispiel von sozialen Stressoren und Gesundheit [Analysis of complex longitudinal data using CTSEM] [Conference presentation]. *13th Meeting of the DGPS Work, Organizational, and Business Psychology Division,* Kassel, Germany.
12) **Irmer, J. P.**, Klein, A. G., & Schermelleh-Engel, K. (2023, March). Estimating power in moderated mediation models and endogenous moderation model: The powerNLSEM R-package [Conference presentation]. *Meeting of the Working Group Structural Equation Modeling 2023 (ATLAS Conference),* Bielefeld, Germany.
11) **Irmer, J. P.**, Klein, A. G., & Schermelleh-Engel, K. (2022, October). Challenges faced in simulation studies examining nonlinear SEM with distributional misspecification [Conference presentation]. *2nd Methods Retreat for Young Researchers,* Kassel, Germany.
10) **Irmer, J. P.**, Klein, A. G., & Schermelleh-Engel, K. (2022, September). Treating ordered-categorical data as continuous in nonlinear SEM: The crucial role of the dependent variable's distribution [Conference presentation]. *52nd Congress of the German Psychological Society (DGPS),* Hildesheim, Germany.
9) Klein, A. G., & **Irmer, J. P.** (2022, September). Causal interpretation of statistical models: Can scientific philosophers give directions for statistical modeling? [Conference presentation]. *52nd Congress of the German Psychological Society (DGPS),* Hildesheim, Germany.
8) **Irmer, J. P.**, Büchner, R. B., & Klein, A. G. (2021, March). A quasi-likelihood ratio test to evaluate heterogeneous growth [Conference presentation]. *1st Methods Retreat for Young Researchers,* Aachen (Online), Germany.
7) **Irmer, J. P.**, Klein, A. G., Gäde, J. C., & Schermelleh-Engel, K. (2021, March). Robustness of LMS compared to regression methods when categorical data are treated as continuous [Conference presentation]. *Meeting of the Working Group Structural Equation Modeling 2021,* Vienna (Online), Austria.

6) **Irmer, J. P.**, Büchner, R. B., Schneider, G., & Klein, A. G. (2019, September). A quasi-likelihood ratio test to evaluate nonlinear SEM using quasi maximum likelihood [Conference presentation]. *14th DGPS Methods & Evaluation Division Meeting,* Kiel, Germany.
5) Holman, D., Escaffi-Schwarz, M., Vasquez, C. A., **Irmer, J. P.**, & Zapf, D. (2019, June). A meta-analytic test of the core mediational proposition of job crafting theory [Conference presentation]. *EAWOP Small Group Meeting "Antecedents of Work Design",* Amsterdam, The Netherlands.
4) **Irmer, J. P.**, Büchner, R. D., & Klein, A. G. (2019, February–March). Global model fit test for nonlinear SEM using quasi maximum likelihood [Conference presentation]. *Meeting of the Working Group Structural Equation Modeling 2019,* Tübingen, Germany.
3) **Irmer, J. P.**, Büchner, R. D., & Klein, A. G. (2018, September). Zur Robustheit des Quasi-Likelihood-Quotienten-Tests für nichtlineare Strukturgleichungsmodelle [On the robustness of quasi-likelihood ratio tests for nonlinear SEM] [Conference presentation]. *51st DGPS Congress,* Frankfurt am Main, Germany.
2) Büchner, R. D., Klein, A. G., & **Irmer, J. P.** (2018, September). Quasi-likelihood ratio test zur Beurteilung der Gesamtmodellgüte von nichtlinearen SEM [A quasi-likelihood ratio test for the evaluation of global model fit of nonlinear SEM] [Conference presentation]. *51st DGPS Congress,* Frankfurt am Main, Germany.
1) **Irmer, J. P.**, Kern, M., Semmer, N. K., & Zapf, D. (2015, September). Meta-analytical results on the challenge-hindrance-stressor concept using ISTA [Conference presentation]. *9th DGPS Work, Organizational, and Business Psychology Meeting,* Mainz, Germany.

Panel Discussions and Symposia

3) Frick, S., **Irmer, J. P.**, Meiser, T., Strobl, C., Siepe, B., & Ulitzsch, E. (2025, September). Quality standards for simulation studies [Panel discussion]. 17th Conference of the Methods & Evaluation Section of the German Psychological Society (DGPs), Berlin, Germany.
2) Gärtner, A., Badstuber, N., Schwake, M., Baierer, F., **Irmer, J. P.**, Henninger, M., Pfeuffer, C., & Ortner, T. (2024, September). Gibt es einen Nachwuchsmangel in der Psychologie? Eine kritische Analyse [Panel discussion]. *53rd Congress of the German Psychological Society (DGPS),* Vienna, Austria.
1) Brandt, H., Hecht, M., Sengewald, M., Frick, S., & **Irmer, J. P.** (2024, September). The use of machine learning for psychological research: What are its opportunities to answer substantive research questions? [Organized symposium]. *53rd Congress of the German Psychological Society (DGPS),* Vienna, Austria.

Events

Workshops

7) Frick, S., Siepe, B., & **Irmer, J. P.** (2026, May). *Wie erforscht man Forschungsmethoden?* [How do we study research methods?]. Online workshop conducted for the Methods and Evaluation Section of the DGPs, Berlin, Germany. Level: M.Sc.
6) **Irmer, J. P.** (2026, April). *Ein bisschen KI schadet nie? – Wie LLMs das Programmieren in R für typische Fragestellungen der Bildungswissenschaften erleichtern* [A little AI never hurt? – How LLMs facilitate R programming for typical questions in educational sciences]. One-day workshop for GRADE, Goethe University Frankfurt. Level: PhD & Post-Doc. 📦 Materials
5) **Irmer, J. P.** (2026, March). *Ein bisschen KI schadet nie? – Wie LLMs das Programmieren in R erleichtern* [A little AI never hurt? – How LLMs facilitate R programming]. Two-half-day workshop for the FDZ Spring Academy 2026, Humboldt-Universität zu Berlin. Level: PhD & Post-Doc. 📦 Materials
4) **Irmer, J. P.**, & Frick, S. (2025, September). *Mathematische Grundlagen von statistischen Tests und Schätzern für Psycholog:innen* [Mathematical foundations of statistical tests and estimators for psychologists]. Full-day workshop at FGME 2025, Germany. Level: PhD. Slides

3) Frick, S., Siepe, B., & **Irmer, J. P.** (2025, May). *Wie erforscht man Forschungsmethoden?* [How do we study research methods?]. Online workshop conducted for the Methods and Evaluation Section of the DGPs, Berlin, Germany. Level: M.Sc.
2) **Irmer, J. P.** (2025, Fall). *Strukturgleichungsmodelle in den Bildungswissenschaften: Komplexere Modellierung latenter Variablen in R* [Structural equation models in educational sciences: Advanced modeling of latent variables in R]. Two-and-a-half-day workshop for GRADE-Education, Goethe University Frankfurt. Level: PhD & Post-Doc.
1) **Irmer, J. P.**, & Schultze, M. (2022, Fall). *Einführung in die Strukturgleichungsmodellierung mit Mplus* [Introduction to structural equation modeling with Mplus]. Two-day workshop for GRADE, Goethe University Frankfurt. Level: PhD & Post-Doc.

Organized Events

Upcoming

2026 **4th Early Career Member Meeting of the Methods and Evaluation Section of the German Psychological Society.** Co-organized with Susanne Frick, Talha Sajjad, and Kai J. Nehler. Three-day meeting in October 2026 at DIPF - Leibniz Institute for Research and Information in Education, in cooperation with Goethe University Frankfurt.

Past

2024 **3rd Early Career Member Meeting of the Methods and Evaluation Section of the German Psychological Society.** Co-organized with Susanne Frick, Florian Scharf, and the organizing team. Three-day meeting held in October 2024 at the University of Kassel, Germany.

Teaching

Colloquium Talks

11) **Irmer, J. P.**, Nehler, K. J., & Schultze, M. (2026, July 7). Can ordered-categorical data ever be treated as continuous? A population-level analysis of the impact of ordinal data on network analysis [Colloquium presentation]. Department of Evaluation, University of Freiburg.
10) Nehler, K. J., **Irmer, J. P.**, & Schultze, M. (2026, June 30). Ordinale Daten als kontinuierlich behandeln: Konsequenzen für die Schätzung psychologischer Netzwerke [Treating ordinal data as continuous: Consequences for the estimation of psychological networks, Department colloquium presentation]. Department of Psychology, University of Erfurt.
9) Nehler, K. J., **Irmer, J. P.**, & Schultze, M. (2026, June 10). Treating ordered-categorical data as continuous: Consequences for sparsity and edge weights in network analysis [Colloquium presentation]. Department of Psychological Methods with Interdisciplinary Focus, Goethe University Frankfurt.
8) **Irmer, J. P.** (2026, June 11). Ein bisschen KI schadet nie? – LLMs, Prompt Engineering und R-Workflows [A little AI never hurts? – LLMs, prompt engineering, and R workflows, Colloquium presentation]. Department of Evaluation, University of Freiburg. Materials
7) **Irmer, J. P.** (2026, May 11). Einführung in Simulationsstudien: Grundlagen und erste Schritte in R [Introduction to simulation studies: Foundations and first steps in R, Colloquium presentation]. Department of Evaluation, University of Freiburg.
6) **Irmer, J. P.** (2026, April 16). Control of baseline confounding in causal panel models: Analytical population-level analyses and simulation results [Colloquium presentation]. Department of Evaluation, University of Freiburg.
5) **Irmer, J. P.** (2026, February 16). Control of baseline confounding in causal panel models: Background and analytical extensions of the Dynamic Panel Model [Colloquium presentation]. Department of Evaluation, University of Freiburg.

4) **Irmer, J. P.** (2025, July 3). Beyond the prompt: Practical AI use in research and teaching [Colloquium presentation]. Department of Psychological Methods, Humboldt-Universität zu Berlin.
3) **Irmer, J. P.** (2025, May 15). From simulation studies to simulation-based power analysis: Tools for modern psychological research [Colloquium presentation]. Department of Psychological Methods, Humboldt-Universität zu Berlin.
2) **Irmer, J. P.** (2024, August 6). My dissertation and beyond [Colloquium presentation]. Department of Psychological Methods, Humboldt-Universität zu Berlin.
1) **Irmer, J. P.** (2022, June 22). Challenges in simulation studies with nonlinear effects, distributional misspecifications, and ordered-categorical data [Colloquium presentation]. Department of Psychological Methods and Evaluation, Goethe University Frankfurt.

Teaching Experience

Courses as Lecturer

Albert-Ludwigs-Universität Freiburg

B.Sc. **Computational Data Analysis.** Seminar; Summer 2026.

M.Sc. **Multivariate Research Methods.** Seminar; Winter 2025.

Humboldt-Universität zu Berlin

B.Sc. **Research Methods II with R.** Seminar; Summer 2024 and Summer 2025.

M.Sc. **Multivariate Research Methods.** Lecture; Winter 2024.

M.Sc. **Multivariate Research Methods with R.** Seminar; Winter 2024.

M.Sc. **Special Topics in Psychology: Developing and Examining Psychological Methods.** Seminar; Summer 2024.

M.Sc. **Special Topics in Psychology: Developing and Examining Psychological Methods Using Artificial Intelligence.** Seminar; Summer 2025.

Goethe University Frankfurt

B.Sc. **Statistics I.** Seminar; Winter 2021 and Winter 2023.

B.Sc. **Statistics II: Advanced Statistics.** Seminar; Summer 2019, Summer 2020, Summer 2021, Summer 2022, and Summer 2023.

M.Sc. **Advanced Research Methods for Psychotherapists.** Seminar; Winter 2021, Winter 2022, and Winter 2023.

M.Sc. **Research Course in Research Methods and Evaluation** on simulation studies and causal modeling of selection effects. Seminar; Winter 2020.

M.Sc. **Research Course in Work and Organizational Psychology** on a meta-analysis on emotional labor. Seminar; Winter 2020.

M.Sc. **Research Methods and Evaluation I.** Seminar; Winter 2019, Winter 2020, and Winter 2021.

M.Sc. **Research Methods and Evaluation II.** Seminar; Summer 2020, Summer 2021, Summer 2022, and Summer 2023.

Courses as Student Teaching Assistant

Goethe University Frankfurt

B.Sc. **Statistics I in Psychology.** Tutorial; Winter 2015.

B.Sc. **Statistics II in Psychology.** Tutorial; Summer 2015 and Summer 2016.

B.Sc. / M.Sc. **Statistics I in Mathematics.** Tutorial; Winter 2015 and Winter 2016.

B.Sc. / M.Sc. **Statistics II in Mathematics.** Tutorial; Summer 2016.

M.Sc. **Practical Course in Python for Psychologists.** Tutorial; Winter 2017.

M.Sc. **Research Methods and Evaluation I.** Tutorial; Winter 2017 and Winter 2018.

M.Sc. **Research Methods and Evaluation II.** Tutorial; Summer 2018.

M.Sc. **Statistics for Environmental Sciences and Biology: Statistics with R.** Tutorial; Winter 2015 and Winter 2016.

Supervision and Methodological Consulting

Bachelor's and Master's Theses
Supervisor or reviewer

2026 **Bachelor's thesis**: *From Functional to Effective Connectivity in Human Intracranial EEG: LMM and Partial Directed Coherence as a Baseline for Dynamic Directional Modeling of Voluntary versus Instructed Hand Movements.*

2025 **Master's thesis**: *Heterogeneity in SEM: Moderated Nonlinear Factor Analysis and Individual Parameter Contribution Regression.*

2023 **Master's thesis**: *Sample Selection Biases and Distributional Misspecifications in the Heckman Model.*

2022 **Bachelor's thesis**: *Scale Shortening Using Artificial Bee Colony Algorithms.*

2022 **Master's thesis**: *Scale Shortening Using Artificial Bee Colony Algorithms.*

2021 **Master's thesis**: *Sample Selection Effects in Mediation Analysis.*

2020 **Master's thesis**: *Scale Shortening Using Genetic Algorithms.*

Statistical and Methodological Consulting

Since 2025 **University of Freiburg — more than 2 consultations.** At least two consultations at professor or postdoctoral level and one at doctoral level.

2024–2025 **Humboldt-Universität zu Berlin — more than 27 consultations.** At least four consultations at professor or postdoctoral level, ten at doctoral level, eleven at master's level, and two at bachelor's level.

2019–2024 **Goethe University Frankfurt — more than 20 consultations.** At least eight consultations at professor or postdoctoral level, four at doctoral level, and eight at master's level.

Academic Service

Academic Service

Since 2025 **Member, Working Group on Preregistration and Quality Standards for Simulation Studies**, Methods and Evaluation Section, German Psychological Society (DGPs).

Since 2023 **Early-Career Researchers' Representative**, Methods and Evaluation Section, German Psychological Society (DGPs). German designation: *Jungwissenschaftler* innenvertretung der Fachgruppe Methoden und Evaluation der Deutschen Gesellschaft für Psychologie*.

2022–2024 **Teaching Staff Representative on the Doctoral Committee**, Institute of Psychology, Goethe University Frankfurt.

Editorial Activities

2023–2026 **Guest Editor**, with P. Schmidt and K. Schermelleh-Engel, for the special issue *Methodological Challenges of Complex Latent Mediator and Moderator Models*, *Behavior Research Methods.* https://link.springer.com/collections/facieeabfj

Reviewing Activities

Journals	*Psychological Methods* · *Behavior Research Methods* · *Multivariate Behavioral Research* · *Journal of Affective Disorders* · *Journal of Cognition*
Conferences	17th Conference of the Methods and Evaluation Section of the German Psychological Society · Early Career Member Meeting of the Methods and Evaluation Section of the German Psychological Society, 2024

Experiences and Skills

Industry Experience

2015	**Research Intern, International Technical Research Center, Adam Opel AG.** Analysis of eye-tracking data from driving-simulator studies to evaluate and improve the usability of in-car infotainment systems.

Technical Skills

Programming and statistical software: R · Python · MATLAB · Mplus · SPSS · Jamovi
Computing infrastructure: Bash · SLURM
Document preparation: LaTeX · R Markdown · Quarto · Microsoft Office
Version control and collaboration: Git · GitHub

Languages

German	Native
English	Fluent
Spanish	Intermediate